AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 5

The AI Conjunction, Debris and Orbital Safety Playbook

How AI watches a crowded sky, prices a close approach honestly, and helps humans decide - without ever mistaking a predicted conjunction for a collision avoided.



Gates CD0-CD5 | Eight templates | 20 golden rules

A predicted conjunction is not a collision avoided. Orbital-safety AI is licensed per object, per shell, per operator and per manoeuvre window - never in general, and never on a probability alone.

CHAPTER 1 - THE ORBITAL SAFETY DECISION BRIEF

A close approach is a decision with a deadline, not an alert with a colour

Volumes 1 and 3 set the standing rule this volume exists to operationalise: AI may screen, rank, draft and recommend, but a human decides every avoidance manoeuvre. This volume licenses the machinery around that decision - the catalogues, the uncertainty, the probabilities, the coordination and the disposal duties - because a good decision made on a bad catalogue is luck wearing a uniform.

Orbital safety is a chain of claims, and each claim has a rung. An object is **tracked**, a pair is **screened**, an event is **assessed**, a manoeuvre is **decided**, and only a practice proven across shells and operators is **transferable**. Most safety failures are rung errors: a screened event treated as assessed, an assessed event treated as decided, a probability treated as a fact.

The licence rule: **orbital-safety AI is licensed per object, per shell, per operator and per manoeuvre window** - never in general, and never on a probability alone.

Standing refusals

- **No autonomous avoidance manoeuvre** outside the fenced envelope of Chapter 11 - and none at all against crewed vehicles, ever.
- **No screening threshold tuned to reduce alert counts** rather than to bound risk; a quieter queue is not a safer orbit.
- **No manoeuvre planned without a secondary-conjunction check** on the post-manoeuvre trajectory.
- **No disposal shortcut:** no passivation skipped, no disposal burn deferred past its window, no dead spacecraft left in a protected shell to save propellant.

EXECUTIVE LENS

A collision is a fleet-ending, market-moving event bought to avoid for the price of analyst hours and delta-v. Fund the chain accordingly.

MISSION LENS

Every gate in this volume protects one decision: move, or do not move, in the window that remains.

OPERATIONS LENS

Staff the assessment desk before you automate it. An unstaffed alert queue is a liability register.

Ten stages from catalogue to disposal, graded stage by stage



Figure 1 - Two spacecraft on converging paths, their uncertainty cones meeting over a debris field: screening sees the geometry; assessment must price it.

The chain below grades each stage for AI licence at CD4, the highest working level short of scale. AI is strongest where the work is volume and pattern - screening, ranking, duplicate detection - and weakest where the output commits propellant, risk or another operator's safety.

Stage	AI licence at CD4	Why
1. Catalogue ingest	Licensed	Format, provenance and staleness checks are mechanical
2. Orbit determination	Licensed, checked	Covariance realism must be independently verified
3. Screening	Licensed	Pure volume problem with testable recall
4. Risk assessment	Licensed, human-reviewed	Probability is an input, not a verdict
5. Operator coordination	Drafting only	Commitments between operators are human acts
6. Manoeuvre planning	Licensed, checked	Options are computed; the choice is not
7. Manoeuvre execution	Refused*	*Except the fenced envelope of Chapter 11
8. Post-event verification	Licensed	Closing the loop is bookkeeping with telemetry
9. Breakup response	Licensed, human-led	Speed matters; so does not manoeuvring into the cloud
10. Disposal watch	Licensed	Deadlines, decay forecasts and passivation checks

Read the map as a budget: licence flows to stages 1 to 4 and 8 to 10 first, because they make the human decision at stages 5 to 7 faster and better informed.

EXECUTIVE LENS

The map shows where the money goes: sensing, data and assessment before autonomy.

MISSION LENS

A stage graded licensed still carries its conditions; the map is a ceiling, not a promise.

OPERATIONS LENS

If a stage has no owner on the roster, its grade is zero whatever the table says.

You cannot avoid what you cannot see, and you cannot see everything

Every orbital-safety claim stands on a catalogue, and every catalogue is incomplete. Public catalogues maintained from government surveillance networks, commercial space situational awareness (SSA) providers, operator ephemerides and an operator's own sensors each see a different sky. The first discipline of this volume is to know which sky each decision was made against.

Catalogue discipline

- **Provenance per object.** Every tracked object carries its source, last observation, fit quality and update cadence. An object whose data is stale is labelled stale in every product that uses it, not silently aged.
- **The lethal non-trackables are named, not forgotten.** Objects too small to catalogue reliably can still end a mission. Shielding, design margins (Volume 2) and shell choice carry that risk; screening cannot, and no dashboard may imply otherwise.
- **Correlations are audited.** AI-assisted track correlation and new-object detection earn their licence on seeded and historical cases, including breakups, deployments and manoeuvring objects that break naive correlation.
- **Manoeuvring counterparties break assumptions.** An object that manoeuvres without notice invalidates predictions made from its old trajectory. The catalogue flags manoeuvrable objects, and assessment treats their covariance accordingly.
- **Own-fleet truth is exact.** The operator's own ephemerides, manoeuvre plans and spacecraft status are shared into the screening chain in full fidelity. More conjunctions are misjudged through stale own-fleet data than through any sensor gap.

Catalogue completeness is reported as a measured quantity per shell - objects tracked against independent estimates - never as an adjective. A safety case that cannot state its catalogue's blind spots has not started.

EXECUTIVE LENS

Catalogue contracts are safety contracts. Buy coverage and cadence, not dashboard polish.

MISSION LENS

Ask each provider what they cannot see, in writing, per shell.

OPERATIONS LENS

Your own data quality is the cheapest catalogue improvement available.

The uncertainty is the product; a precise wrong orbit is worse than an honest vague one

A conjunction prediction is two trajectories and two uncertainty volumes arriving at the same place at the same time. The trajectories get the attention; the covariances decide the answer. This chapter treats uncertainty quantification as the core product of the safety chain.

- **Covariance realism is tested, not assumed.** Predicted uncertainties are scored against later, better observations across thousands of objects. A system whose 3-sigma volumes contain reality half the time is recalibrated or its outputs are widened before use - automatically.
- **Staleness inflates, visibly.** As an ephemeris ages, its covariance grows along a published curve per object class. Products show data age beside every risk figure.
- **Drag is the great confounder.** Atmospheric density varies with solar and geomagnetic activity, and low-orbit predictions inherit that weather. During geomagnetic storms, assessment widens uncertainties, shortens decision horizons and says so in the event record.
- **AI orbit determination is checked classically.** Learned orbit-determination and propagation aids are validated against physics-based propagators on held-out objects; divergence beyond tolerance demotes the learned aid to advisory.
- **Manoeuvre detection is a safety function.** Detecting that a counterparty has manoeuvred - and voiding predictions based on its prior state - is licensed early, because it prevents confident assessments of a sky that no longer exists.

The test of this chapter is simple: pick any closed event and ask what the system believed, when, and how reality scored against that belief. If the chain cannot answer, its probabilities are decoration.

EXECUTIVE LENS

Fund covariance validation like it is the product, because it is.

MISSION LENS

A vendor who will not show calibration curves has shown you something else.

OPERATIONS LENS

Storm clauses belong in the safety case before the storm, not during it.

Millions of pairs, a handful of decisions, and no permission to miss quietly

Screening compares every relevant object pair across the planning horizon and surfaces the fraction that deserve human eyes. At constellation scale this is the largest computation in the safety chain and the natural home of AI - provided its licence is earned on recall, not on queue size.

- **Recall is the licence metric.** Screening is tested against independently computed reference screenings and historical events. A missed event that reference screening catches is a licence event: investigated, disclosed in the safety log, and counted.
- **Thresholds are set by risk appetite, signed.** Screening volumes and probability floors are set in the safety case and changed only through it. Tuning a threshold because the queue is long is a refusal under Chapter 1.
- **Duplicates and re-alerts are merged, not stacked.** The same event arriving from three catalogues is one event with three witnesses. AI deduplication earns real analyst hours here - this is where most honest savings in the worked example live.
- **Workload is forecast, not discovered.** Screening predicts event volume days ahead by shell and counterparty class, so assessment staffing follows the sky rather than being ambushed by it.
- **Crewed and high-value objects screen separately.** Events involving crewed vehicles or designated high-value assets bypass all triage economies and go straight to human assessment, always.

Screening success is not a short queue. It is a queue whose every entry deserves to be there and whose absences can be defended, event class by event class, under audit.

EXECUTIVE LENS

Judge screening vendors on audited recall, never on alerts suppressed.

MISSION LENS

Alert volume is a staffing forecast, not an embarrassment to be tuned away.

OPERATIONS LENS

The events you never saw are the metric. Fund the reference checks that find them.

A probability is a model of your ignorance, priced in someone else's spacecraft

Collision probability (Pc) is the most quoted and most misused number in orbital safety. It depends on miss distance, combined hard-body radius, and covariances that Chapter 4 grades as honest or not. This chapter fixes how Pc may be computed, presented and acted on.

- **Pc dilution is named on the page.** As covariance grows, Pc for a given geometry can fall - uncertainty can make an event look safer while making knowledge worse. Every assessment shows Pc alongside covariance quality and data age; a falling Pc driven by growing uncertainty triggers more tracking, not relief.
- **Maximum Pc bounds the ignorance.** Where covariance is suspect, assessments also compute the maximum achievable Pc over plausible uncertainties. Decisions use the bound, not the point estimate, when the two disagree materially.
- **Risk is probability times consequence.** A 1-in-10,000 chance against a derelict rocket body and the same chance against a cubesat are different decisions: mass, fragment yield, shell persistence and counterparty manoeuvrability all enter the risk record.
- **Cumulative risk is budgeted.** A fleet that accepts hundreds of individually tolerable events a year has made a portfolio decision. Annual accepted-risk budgets per shell make that decision explicit and owned (Chapter 13).
- **Thresholds are decision rules, not physics.** Act / monitor / dismiss thresholds are policy, set by the accountable operator, recorded per event class - and never reverse-engineered from how busy the desk is.

House rule: **no Pc is quoted without its covariance quality, its data age and its maximum-Pc bound.** A naked probability is returned to sender.

EXECUTIVE LENS

Boards should see cumulative accepted risk, not a count of alerts closed.

MISSION LENS

Pc is an argument, not an answer. Fund the people who can cross-examine it.

OPERATIONS LENS

Consequence belongs in the record: fragment yield outlives the event.

The event file is the decision's memory; write it like it will be read after an accident

Assessment turns a screened pair into an event file: geometry, timing, uncertainties, counterparty identity, communications, options and a recommendation. Conjunction Data Messages (CDMs), in the standard CCSDS format, are the shared currency between providers and operators; this chapter governs how AI assembles, checks and learns from them.

- **Every event has one file and one owner.** AI assembles the file from all CDM updates, own-fleet state and tracking tasking; a named analyst owns it until closure.
- **Updates re-open, never overwrite.** Each new CDM supersedes visibly, with the risk trajectory shown over time. An event whose risk is climbing as time-to-closest-approach (TCA) shrinks follows the escalation clock, automatically.
- **Tracking is tasked, not hoped for.** High-risk events generate sensor tasking requests to shrink covariance while shrinking it still helps. The value of information is computed per event: some uncertainties are worth money to reduce, others are not.
- **Decision deadlines are derived, not felt.** The latest decision time follows from manoeuvre planning lead time, pass availability (Volume 3) and counterparty coordination time - computed per event and displayed as a countdown the whole desk can see.
- **Closure requires verification.** An event closes when post-TCA observations confirm the outcome or a manoeuvre's execution is verified - not when TCA passes in silence (Chapter 9).

After any collision or near-miss inquiry, this file is what the operator actually knows. AI's job is to make it complete, contemporaneous and impossible to tidy up after the fact.

EXECUTIVE LENS

The event file is your evidence and your memory. Its completeness is a safety metric.

MISSION LENS

Escalation clocks beat escalation instincts, especially at 3 a.m.

OPERATIONS LENS

Close events on evidence. Silence past TCA is an open event wearing a costume.

Somebody has to move; deciding who is a negotiation, not a computation



Figure 2 - The flight dynamics officer and the screening intelligence meet at the amber decision ring: the system proposes, the human decides.

When both objects are alive and operated, the safest outcome is usually one agreed manoeuvre, not two improvised ones. Coordination is where orbital safety becomes diplomacy conducted against a countdown, and where AI helps most by drafting, translating and remembering - never by committing.

- **Who moves is agreed in advance where possible.** Standing coordination agreements - by shell, object class and manoeuvre capability - are negotiated before events, with operators of recurring counterparties. AI maintains the register and surfaces the applicable agreement per event.
- **Contactability is a safety property.** Operators publish and maintain monitored conjunction contacts. An uncontactable operator is a risk fact recorded per counterparty and priced into shell decisions (Chapter 13).
- **Silent and dead counterparties change the maths.** Against a derelict, the whole burden is yours and the fragment consequence is higher. Assessment states counterparty status in its first paragraph, not its footnotes.
- **Both-move is a designed-out failure.** Uncoordinated simultaneous manoeuvres can re-create the conjunction they fled. Commitments, once exchanged, lock: the coordinating operator stands still unless the agreement is formally revised.
- **Crewed vehicles are absolute.** Events involving crewed stations follow the station's own protocols and timelines, which override every economy in this volume. The other party moves, early, and without negotiation about it.

AI drafting earns its keep in the small hours: correct contacts, current agreements, proposed windows and a clean record of who agreed what, in the formats each counterparty actually reads.

EXECUTIVE LENS

Coordination agreements signed in peacetime are worth more than autonomy bought for the crisis.

MISSION LENS

Publish your conjunction contact and staff it. Silence is a risk you export to everyone else.

OPERATIONS LENS

Record counterparty reliability. The register will change how you choose shells.

Moving is easy; moving without creating the next event is the job

Avoidance manoeuvres are small burns with large paperwork. Planning computes options that reduce risk below threshold while respecting propellant budgets, mission impact, spacecraft constraints from Volume 3 - and the obligation to check what the new trajectory meets next.

- **Options, plural, with prices.** The planner presents at least two feasible options where physics allows, each priced in delta-v, mission interruption, propellant-life impact and residual risk. A single take-it-or-leave-it option is a decision made by the software wearing the planner's clothes.
- **Secondary screening is mandatory.** Every candidate post-manoeuvre trajectory is screened across the full horizon before the option may be recommended. A manoeuvre that resolves today's event by creating Thursday's is a failed plan, however elegant.
- **Return is planned with departure.** Station-keeping return, phasing recovery and service resumption are part of the same plan, priced before approval, so the decision sees the whole cost.
- **Constraints are hard, preferences are soft.** Eclipse, power, thermal, ground-contact and payload commitments enter as constraints from Volume 3's operating picture, not as surprises discovered after the burn.
- **Execution is verified.** Post-burn tracking confirms the achieved trajectory; deviation beyond tolerance re-opens the event and triggers re-screening on the as-flown orbit.

Planning quality is measured after the fact: predicted versus achieved miss distance, residual risk, and secondary events created. The planner - human or machine - is graded on its record, per shell.

EXECUTIVE LENS

Price every option in delta-v and fleet life, where the decision can see it.

MISSION LENS

The secondary check is not bureaucracy. It is the difference between avoidance and displacement.

OPERATIONS LENS

Grade planners on as-flown outcomes, not on plans submitted.

A named human, a recorded rationale, and no decision by default

Volumes 1 and 3 placed the manoeuvre decision with a human. This chapter builds the room where that decision happens: who holds authority at each risk level, what they must see, and how the decision is recorded so that deciding nothing is also a decision someone owns.

Event class	Decision authority	Minimum record
Routine, below action threshold	Duty analyst	Event file, risk trajectory, dismissal rationale
Action threshold crossed	Duty flight dynamics lead	Options with prices, coordination record, decision and time
High consequence (mass, fragments, shell)	Mission director on call	Full file, maximum-Pc bound, cumulative-risk impact
Crewed vehicle involved	Per station protocol	Station timeline compliance, early-move evidence
Decision deadline reached undecided	Escalates automatically	Non-decision recorded as a decision, with owner

- **Recommendations are labelled as recommendations.** AI output in the file carries its rung: screened, assessed, or recommended. Nothing in the file may wear a higher rung than it earned.
- **Dissent survives.** An analyst who disagrees with the decision records it in the file. Post-event reviews read the dissents first.
- **Fatigue is a safety input.** Decision authority rotates with protected rest; the escalation clock assumes a human who is awake enough to be accountable (Volume 3 staffing discipline, applied to safety).

EXECUTIVE LENS

Authority tables written in calm are what stand up in crisis. Write yours now.

MISSION LENS

A non-decision is a decision. Make the record say whose.

OPERATIONS LENS

Protect the decider's sleep like you protect propellant.

The narrow envelope where a machine may move a spacecraft to save it

There is one honest case for autonomous avoidance: events that mature faster than humans can be reached - short-warning conjunctions against well-tracked objects, in shells where the operator has pre-decided its risk policy in detail. This chapter licenses that case narrowly, and refuses everything wider.

The fence - all conditions, all the time

- **Pre-authorised envelope only.** Autonomous action is limited to a signed envelope: object classes, shells, risk thresholds, maximum delta-v per event and per quarter, and allowed manoeuvre directions computed and secondary-screened in advance.
- **Crewed and high-consequence exclusions are absolute.** No autonomous action on events involving crewed vehicles, large derelicts, or any counterparty whose status is uncertain.
- **Every autonomous action is reported immediately** and reviewed by a human within one orbit period; two anomalous autonomous actions suspend the envelope pending review.
- **The envelope expires.** Re-authorisation is a fresh decision on fresh evidence, per shell, at most annually - never a renewal by inertia.
- **Refused outright:** autonomous avoidance as a substitute for staffing the assessment desk; autonomous action based on a single catalogue source; any learning-in-flight change to the envelope itself.

The fence rule: **autonomy may execute a decision the operator has already made in general; it may never make the operator's decision in particular.**

EXECUTIVE LENS

If you cannot staff the desk, fix the roster before you widen the fence.

MISSION LENS

An envelope that never expires is not an envelope.

OPERATIONS LENS

Autonomous actions get reviewed like manoeuvres by a new hire: every one, at first.

When an object becomes a cloud, the catalogue, the plan and the week all change

Breakups - collisions, explosions of stored energy, deliberate destructions - convert one tracked object into hundreds or thousands of fragments in minutes. The response is a rehearsed surge, not an improvisation, and AI's value is speed of characterisation under human direction.

- **Detect and declare.** Breakup detection fuses sensor reports, catalogue anomalies and operator notifications. Declaration starts the clock and the playbook; arguing about certainty while fragments disperse is the known failure.
- **Characterise the cloud.** Fragment count estimates, energy distribution, cloud evolution modelling and shell-crossing forecasts are produced with their uncertainties stated. Early models are wrong in known directions; the record keeps each revision.
- **Re-screen everything, in priority order.** The fleet and affected shells are re-screened against the evolving cloud model, crewed and high-value assets first. Manoeuvring into a poorly characterised cloud is a named risk the mission director must accept explicitly.
- **Own-fleet breakups carry extra duties.** If the fragments are yours: notification, data sharing, root-cause investigation and regulator engagement follow Chapter 18, and your own event files are preserved unedited for it.
- **Attribution is careful and slow.** Cause assessment uses preserved evidence and independent review. Public attribution on model output alone is refused; the diplomatic and legal consequences outlive the news cycle.

Breakup drills run at least annually with a simulated cloud, including the communications surge. The first real breakup an organisation meets should not be the first one it has ever practised.

EXECUTIVE LENS

Rehearse the surge. Breakups do not schedule themselves around your staffing.

MISSION LENS

Cloud models are revised in public, inside the record. Certainty theatre helps no one.

OPERATIONS LENS

If it is your breakup, your duties double and your editing rights end.

Every shell has a budget; the fleet plan is where it gets spent

Orbital safety scales from events to environment. Shells have finite carrying capacity: beyond a population density, collision-generated fragments outpace removal and decay, and the shell degrades for everyone - the dynamic behind the Kessler syndrome scenario. This chapter makes the environment a managed quantity, not a backdrop.

- **Shell budgets are explicit.** Per shell: object population, your share, accepted annual collision risk, fragment-generation budget and disposal reliability - reviewed annually against environment models.
- **Environment models are inputs, with names.** Long-term evolution models used for planning are identified, versioned and challenged; AI-accelerated modelling is validated against the reference models it accelerates.
- **Your footprint includes your dead.** Derelicts you own count against your budget at full consequence until they decay or are removed. The register of own derelicts is a board-visible number.
- **New deployments argue with the budget.** Shell selection for new missions (Volume 2) presents the environmental cost alongside performance: density, persistence at altitude, and the fragment yield if the worst happens.
- **Collective action is a strategy, not a hope.** Data sharing, coordination norms and removal funding (Chapters 16 and 19) are how individual budgets stay meaningful in a shared shell.

An operator who can state its shell budgets and its derelict count in one breath is managing the environment. Everyone else is renting luck from it.

EXECUTIVE LENS

Ask for the derelict register. If there is not one, that is the finding.

MISSION LENS

Shell choice is a decades-long commitment priced in other people's risk too.

OPERATIONS LENS

Carrying capacity turns sustainability from a value into a number. Use the number.

The cheapest fragment is the one the spacecraft was built never to make

Mitigation is decided years before the event, in design reviews governed by Volume 2. This chapter lists the design duties the safety chain audits, aligned with international debris-mitigation guidelines and the licensing conditions that increasingly enforce them.

- **Passivation is non-negotiable.** Stored energy - propellant, pressure, batteries, momentum devices - is depleted or safed at end of mission, by design that works even if the spacecraft is dying. Passivation success is verified by telemetry where possible, and the verification is filed.
- **Breakup prevention in life, too.** Design and operations limit accidental fragmentation risk during the mission: shielding for critical elements, battery and tank management, and anomaly responses that safe the vehicle early (Volume 3).
- **Design for demise and for capture.** Materials and structures chosen to burn up on re-entry (Chapter 17), and - for large or long-lived objects - grapple features and markers that keep future removal physically possible (Chapter 16).
- **Release nothing casually.** Mission-related objects, lens caps, deployment mechanisms and separation debris are inventoried, minimised and tracked as your own catalogue entries from day one.
- **AI's role is audit.** Design-stage AI checks compliance evidence, hunts missing passivation paths in fault trees, and flags components whose failure modes create fragments. It does not certify; accountable engineers certify.

Every derelict in today's catalogue was once a design review that accepted a risk. This chapter exists so the acceptance is conscious, priced and rare.

EXECUTIVE LENS

Passivation evidence belongs in the delivery paperwork, not in a promise.

MISSION LENS

Audit the fault trees for fragment-making failures specifically. They hide in other people's chapters.

OPERATIONS LENS

Design for capture is cheap now and priceless in twenty years.

A mission is not over when the revenue stops; it is over when the object is safe

Disposal is where orbital safety meets contract law and conscience. Post-mission disposal - deorbit from low orbit within the period your licence requires (five years under current United States rules for many operators, tightened from the older 25-year guideline), or transfer to a graveyard orbit from geostationary altitude - must succeed even when the spacecraft is old, degraded and flown by a skeleton team.

- **Disposal reliability is a number with evidence.** The probability that disposal will succeed is estimated at design, tracked through life as health changes, and reported per spacecraft. A fleet average that hides three unreliable vehicles repeats Volume 3's dashboard failure in its most permanent form.
- **Propellant and health are reserved, not hoped for.** Disposal delta-v is ring-fenced from mission use; end-of-life health reviews begin years before the window, while options still exist.
- **Disposal is verified.** Graveyard insertion or re-entry is confirmed by tracking, and the confirmation closes the object's file. Attempted is not a status; achieved is.
- **Failure has a plan.** A spacecraft that cannot dispose itself enters the derelict register with its risk assessed, counterparties informed through the catalogue, and removal options (Chapter 16) evaluated - not quietly forgotten.
- **Disposal watch is licensed AI work.** Decay forecasting, disposal-window tracking and passivation verification monitoring run continuously, escalating on slippage while correction is still cheap.

Disposal rule: **the licence to occupy a shell is rented, and the rent is a verified exit.**

EXECUTIVE LENS

Ring-fence disposal propellant in the budget system, not in a slide.

MISSION LENS

Per-spacecraft disposal reliability, or it did not happen.

OPERATIONS LENS

A failed disposal is a disclosed event with a plan, never a quiet line in a register.

Taking hold of a tumbling rocket body is rendezvous with an adversary that is only physics

Active debris removal (ADR) - capturing and deorbiting dead objects - is where orbital safety becomes a flight programme of its own, importing every discipline from Volumes 1 to 3 plus three that are unique: the target did not consent, cannot cooperate, and may belong to someone else.

- **Consent and title are gates, not paperwork.** Objects remain under their owner's jurisdiction. Removal proceeds only with the owner's agreement and the licensing state's authorisation, documented before rendezvous - whatever the target's condition.
- **Characterisation before commitment.** Tumble state, structure, stored energy and grapple suitability are observed and modelled over time; AI perception aids earn their licence on synthetic and analogue targets before flight targets (Volume 2's twin discipline, at closing velocity).
- **Abort is designed at every range.** Approach corridors carry hold points and retreat trajectories that are themselves conjunction-screened. A removal vehicle that becomes debris has failed at the one job it had.
- **Dual-use sensitivity is managed openly.** Rendezvous and capture capability is inherently dual-use. Transparency measures - published mission profiles, observation sharing, notified windows - are part of the mission design, not public relations added later.
- **Economics are honest.** Removal costs per object are compared against risk retired, per shell, and the target list is prioritised by risk - mass, persistence, fragment yield - not by convenience or publicity.

Licensed narrowly, ADR is the only tool that reduces the existing derelict stock. Licensed loosely, it is an uncrewed grappling programme near other people's assets. The CD gates apply with their evidence doubled.

EXECUTIVE LENS

Fund characterisation time. Tumbling targets punish schedules.

MISSION LENS

Consent first, rendezvous second - in the documents and in fact.

OPERATIONS LENS

Judge ADR by risk retired per shell, not objects removed per press release.

The last conjunction is with the ground, and the public never signed the safety case

Disposal by re-entry trades orbital risk for ground risk, and the trade is governed by casualty-expectation standards that the operator owes to people who never chose to participate. AI supports the prediction; the accountability is not delegable.

- **Casualty expectation is computed per event.** Surviving debris footprints, population exposure and casualty expectation are calculated with current models for every controlled and significant uncontrolled re-entry, and recorded against the applicable standard.
- **Design for demise is verified, not asserted.** Demise modelling is validated against test and flight evidence per spacecraft family; components predicted to survive are listed, with their materials and masses, in the safety file (Chapter 14).
- **Controlled re-entry is preferred where risk demands.** Large objects target remote ocean areas with the manoeuvre chain of Chapter 9 and the verification of Chapter 15. The delta-v to do this is reserved at design, not negotiated at end of life.
- **Prediction uncertainty is communicated honestly.** Re-entry windows narrow late and footprints move. Public and authority communications state uncertainty as ranges and update visibly; false precision about impact points is a refusal.
- **AI break-up and survivability models are benchmarked** against the reference tools and test data of the field, per vehicle family, before their outputs support a licence filing.

Orbital safety ends at the surface. A programme that is meticulous at 550 kilometres and casual about the last 100 has misunderstood whose risk it is managing.

EXECUTIVE LENS

Casualty expectation is a number owed to strangers. Compute it like it matters, because it does.

MISSION LENS

False precision in public re-entry statements is a trust debt with interest.

OPERATIONS LENS

Reserve controlled-re-entry propellant at design review.

The legal sky: registration, liability and the licence conditions that outlive the mission



Figure 3 - The Aurora orbital safety wall: every shell under watch, and one derelict rocket body held in red quarantine.

Orbital safety operates inside a legal framework built from the Outer Space Treaty and the Liability and Registration Conventions, national licensing, and regulator conditions that have tightened steadily - including post-mission disposal timelines, conjunction-data sharing duties in some jurisdictions, and casualty-risk standards for re-entry.

- **Liability follows the launching state and the operator.** Damage on the ground or to aircraft, and fault-based liability in orbit, create exposures that event files (Chapter 7) will one day be used to judge. Documentation quality is a legal control, not an administrative taste.
- **Licence conditions are tracked as obligations,** per spacecraft, in a register the safety chain reads: disposal deadlines, reporting duties, coordination requirements. A condition discovered at its deadline is a governance failure the register exists to prevent.
- **Insurance prices behaviour.** Underwriters increasingly read disposal reliability, coordination records and near-miss history. The safety metrics of this volume are becoming premium inputs; managing them honestly is cheaper than performing them.
- **Incident duties are rehearsed.** Collision, breakup and casualty events trigger notification, preservation and cooperation duties across jurisdictions. The contact tree and evidence-preservation steps are drilled with the breakup playbook (Chapter 12).
- **AI advises; counsel decides.** Regulatory AI tools may track obligations and draft filings, but positions taken with regulators carry human legal sign-off, per filing, without exception.

Regulation in this domain moves in one direction. Programmes built to today's floor will retrofit at tomorrow's prices; programmes built to this volume's standard mostly will not.

EXECUTIVE LENS

The obligation register is a flight system. Treat missed conditions like missed manoeuvres.

MISSION LENS

Your event files are tomorrow's evidence. Write them for the inquiry you hope never happens.

OPERATIONS LENS

Insurance is becoming an audit. Keep the metrics real.

Safety data hoarded is safety halved; the commons is an operating system, not a slogan

No operator sees the whole sky. Conjunction safety depends on shared catalogues, operator ephemeris sharing, standard message formats and emerging civil space-traffic coordination services - including the United States' Traffic Coordination System for Space (TraCSS) programme and international counterparts. This chapter sets the sharing duties of a licensed operator.

- **Share your best data, on time.** Own-fleet ephemerides with covariance, manoeuvre plans and spacecraft status are contributed at the fidelity you would want from counterparties. Withholding precision you possess while screening on others' data is a named integrity failure.
- **Standards are followed exactly.** CDM and ephemeris exchange use the published CCSDS standards without local dialect; interoperability failures in this chain surface as missed events, not as IT tickets.
- **Shared data is still graded.** Incoming data carries provenance and quality per Chapter 3; a coordination service's output enters assessment as evidence, not as verdict - accountability for the decision stays with the operator.
- **Security shapes, but does not silence, sharing.** Sensitive fleets share through the protected channels their regulators provide. The existence of a security constraint is managed with the regulator, not used as a general reason to go dark.
- **AI interoperability is tested.** Screening and assessment tools are exercised against partner data in joint drills, including format edge cases, before an event depends on them.

The commons framing is practical, not idealistic: your fleet's survival literally depends on the quality of data other operators volunteer. Contribute like it.

EXECUTIVE LENS

Your ephemeris-sharing quality is your counterparties' safety margin. Fund it as such.

MISSION LENS

Accountability cannot be outsourced to a coordination service. The decision stays yours.

OPERATIONS LENS

Drill the interfaces. Format failures are conjunction risks with paperwork.

Buying awareness is easy; knowing what your awareness is worth is the discipline

Commercial SSA - tracking, catalogue services, screening platforms and analytics - has made sophisticated awareness buyable. This chapter applies Volume 1's vendor discipline to the safety chain, where a vendor's error bars become your fleet's risk.

- **Evaluate on your sky.** Vendor screening and assessment are scored against your fleet's historical and seeded events, in your shells, before contract - including their covariance calibration curves (Chapter 4), not their brochure accuracy.
- **Provenance and sensor mix are disclosed.** Which sensors, which tasking priority your contract buys, what happens to your coverage when a larger customer has a crisis - in the contract, not in the sales deck.
- **Sovereign and allied sources are part of the design.** Government catalogues and allied sharing provide a baseline no commercial contract should silently replace. The safety case names which sources are authoritative for which decision class.
- **Exit and continuity are safety properties.** Catalogue history, event files and calibration records are exportable in standard formats, tested annually. Losing your safety memory at a contract break is an operational risk, priced as one.
- **Vendor models update under change control.** A silently updated screening model is Volume 3's fleet-update failure in a new uniform: version, validation evidence and rollback path accompany every change, or the change does not ship.

The worked example that follows shows a competent vendor failing these tests in instructive ways - and the parts of its service that earned a licence anyway, once the claims were priced honestly.

EXECUTIVE LENS

Score vendors on your historical events before signature. Afterwards is advocacy.

MISSION LENS

Change control on safety models is non-negotiable, whoever owns the model.

OPERATIONS LENS

Test the exit while you still like each other.

Delta-v is money, analyst hours are money, and a collision is all the money

The unit metric of this volume is **cost per verified resolved conjunction**: the fully loaded cost of the safety chain - sensors and data, software, assessment staffing, coordination, manoeuvre propellant priced as fleet life, and mission interruption - divided by events verifiably resolved at or below the accepted risk threshold. It disciplines both directions of waste: under-spending that leaves risk unassessed, and over-manoeuvring that spends fleet life on noise.

- **False manoeuvres are costs, not caution.** Every unnecessary burn spends propellant-life and creates secondary-screening load. Manoeuvre precision - fraction of executed manoeuvres that post-analysis confirms were needed - is reported beside the unit cost.
- **Missed assessment is the unpriced catastrophe.** The metric never rewards a smaller queue. Recall failures (Chapter 5) are reported in the same table as costs, so cheapness bought with blindness is visible at a glance.
- **Claimed, released, banked.** Vendor savings claims are audited in Volume 1's three columns: hours claimed, hours verifiably released from rosters, hours banked as budget. Automation that moves work from analysts to exception-handling has moved the cost, not removed it.
- **Delta-v is accounted as currency.** Annual avoidance delta-v per spacecraft, priced against fleet-life extension value, appears in the same report as cash costs - because to the fleet, it is cash.
- **Disposal and removal are in the same ledger.** Mitigation spend is compared against risk retired (Chapters 13 and 16), so prevention, avoidance and removal compete honestly for the safety budget.

EXECUTIVE LENS

Demand claimed, released and banked in three columns, from every vendor, every time.

MISSION LENS

Report recall next to cost. A cheap blind system is the most expensive kind.

OPERATIONS LENS

Propellant spent on noise is fleet life burned for comfort. Track who approved it.

Anyone can halt the automation; restarting it takes evidence and two signatures

The operating model closes the volume: the people, rosters and authorities that make the previous twenty-one chapters a practice rather than a position.

- **The safety desk is a licensed function.** Assessment authority is held by named, trained analysts per shift, with the decision table of Chapter 10 posted and drilled. Unfilled shifts demote AI licences automatically - screening may run, assessment recommendations may not issue.
- **Anyone can halt.** Any operator may suspend an AI safety service - screening automation, planning aids, the Chapter 11 envelope - on a stated safety concern, without permission and without penalty. The halt is logged and reviewed, never relitigated in the moment.
- **Restart takes two signatures,** one from flight dynamics and one from mission leadership, on evidence that the halt's cause is understood and addressed. Restart on a timer is prohibited.
- **Near-miss review is routine.** Weekly review of the closest events, the dissents (Chapter 10) and the calibration record; monthly review of cumulative risk against budget (Chapter 13); annual re-licensing of every AI service in the chain.
- **The profession is built deliberately.** Flight dynamics and SSA analysis are crafts with apprenticeships. AI that drafts, ranks and remembers should make analysts better faster - organisations audit whether it is doing that, or quietly deskilling the desk it protects.

Orbital safety is a promise made by people, kept with machines. The model above keeps the promise's authorship where the liability - and the credit - belongs.

EXECUTIVE LENS

Halt authority without penalty is the cheapest safety device in this volume. Install it.

MISSION LENS

Two signatures to restart, zero to halt. Asymmetry is the design.

OPERATIONS LENS

Audit deskilling annually. The desk is the fallback for everything else in this book.

WORKED EXAMPLE - AURORA ORBITAL, PART 1

The audit: a fleet-wide safety claim meets its own event files

All figures in this worked example are illustrative. Aurora Orbital (Volumes 1 to 4) is offered a safety platform claiming to cut conjunction-management cost by **52 percent**, release **58,000 analyst hours** a year, screen with effectively perfect recall, and execute autonomous avoidance for the fleet - at decision-ready grade, across all shells, from day one.

What the audit found

- **Recall was 96.8 percent, not perfect** - against independently computed reference screening, the platform missed 41 historical events, 3 of which had crossed the action threshold. Every miss traced to manoeuvring counterparties correlated onto stale tracks (Chapter 3).
- **Its probabilities were overconfident.** Covariance calibration failed Chapter 4's test in the two highest shells: stated 3-sigma volumes contained reality 71 percent of the time. Several dismissed events carried maximum-Pc bounds above the action threshold.
- **38 percent of its alert volume was duplicates** across catalogue sources - the honest saving, once deduplication was isolated and priced.
- **One proposed autonomous manoeuvre created a secondary conjunction** with a derelict rocket body inside 72 hours, caught by Aurora's own mandatory secondary check (Chapter 9), which the platform performed only on request.
- **Disposal reliability was reported as a fleet average of 96 percent** - hiding three spacecraft at or below 70 percent, two already inside their disposal-planning windows.

The claim collapsed in the usual order: autonomy first, then the probabilities, then the hours - leaving a competent screening and workflow core that had never needed the inflation.

EXECUTIVE LENS

Audit recall before price. The misses set the ceiling on everything else.

MISSION LENS

Calibration curves are the vendor's real brochure. Read those.

OPERATIONS LENS

A fleet average hiding three unreliable disposals is the audit in one line.

WORKED EXAMPLE - AURORA ORBITAL, PART 2

CD5 granted in part: the licence that survived was smaller and worth more

All figures illustrative. Priced honestly, the platform's 52 percent saving became **16 percent verified**: **24,700 hours released** from rosters - mostly deduplication, event-file assembly and coordination drafting - of which **14,900 banked** as budget, the rest reinvested in assessment staffing and covariance validation.

Licensed at CD4

- Screening with audited recall of 99.5 percent on Aurora's shells after the manoeuvring-object correlation fix, with quarterly reference checks as a licence condition.
- Event-file assembly, CDM management, escalation clocks and coordination drafting.
- Manoeuvre option computation with mandatory secondary screening built in, and as-flown grading of every plan.
- Disposal watch with per-spacecraft reliability reporting; the three unreliable vehicles entered funded remediation plans.

Refused

- Autonomous avoidance execution - the Chapter 11 envelope was not granted; the secondary-conjunction failure ended the application for this cycle.
- Probability-only dismissal of events whose maximum-Pc bounds crossed threshold.
- Fleet-average disposal reporting, in any product, ever again.

Cost per verified resolved conjunction in licensed cells: **0.85 against a 0.98 baseline**. CD5 scale was granted in part - screening and workflow across the fleet, assessment support per shell as calibration is proven - and the refusal list was published beside the licence, where the next vendor could read it.

EXECUTIVE LENS

A smaller true licence beats a grand false one, in safety above all.

MISSION LENS

Publish refusals beside licences. The market learns, and so does the next audit.

OPERATIONS LENS

Savings reinvested in assessment staffing are still savings. They are just honest ones.

THE ORBITAL SAFETY STANDARD

One page, to be read before any manoeuvre window opens

1. Every claim carries its rung - tracked, screened, assessed, decided, transferable - earned per object, per shell, per operator and per manoeuvre window.

2. A human decides every manoeuvre. Automation recommends, drafts and remembers; the Chapter 11 fence is the only exception, and it executes prior human decisions.

3. No probability is quoted without covariance quality, data age and the maximum-Pc bound.

4. Screening thresholds are set by signed risk appetite and never tuned to shrink a queue.

5. Every manoeuvre plan is secondary-screened, priced in delta-v and fleet life, and graded as flown.

6. Coordination commitments, once exchanged, lock. Both-move is designed out, not apologised for.

7. Events close on post-event verification, not on silence past TCA.

8. Disposal is verified per spacecraft; passivation is evidenced; derelicts are registered and board-visible.

9. Own-fleet ephemerides and manoeuvre plans are shared at full fidelity, on time, in standard formats.

10. Anyone can halt the automation. Restart takes evidence and two signatures.

If an event file cannot state its rung, its uncertainty and its decision owner in one sentence, **the event is not ready to be decided.**

EXECUTIVE LENS

Print it where the night shift can see it.

MISSION LENS

Ten clauses, each bought by somebody's near miss.

OPERATIONS LENS

The one-sentence test closes most arguments before TCA does.

SIX FAILURE PATTERNS

How orbital safety programmes actually go wrong

1. The tuned-away queue

Alert volume embarrasses someone senior; thresholds rise; the queue shortens and the first missed event arrives on schedule. Antidote: thresholds signed in the safety case (Chapters 5 and 6).

2. The confident covariance

Probabilities are computed on uncertainties nobody validated; dilution makes growing ignorance look like falling risk. Antidote: calibration testing and maximum-Pc bounds (Chapters 4 and 6).

3. The helpful double move

Both operators manoeuvre, uncoordinated, and meet again on the far side. Antidote: standing agreements and commitment locks (Chapter 8).

4. The manoeuvre into Thursday

Today's avoidance creates a worse event inside the week, discovered by someone else's screening. Antidote: mandatory secondary screening (Chapter 9).

5. The averaged derelict

Fleet disposal reliability reads 96 percent while three spacecraft cannot dispose at all. Antidote: per-spacecraft reporting and the derelict register (Chapters 13 and 15).

6. The silence after TCA

Events close because nothing was heard, and the near miss that did happen is never learned from. Antidote: closure on verification only (Chapter 7).

EXECUTIVE LENS

Each pattern was a shortcut that looked like efficiency.

MISSION LENS

Patterns 2 and 5 are honesty failures wearing statistical clothes.

OPERATIONS LENS

Teach the patterns; desks spot what they can name.

30 QUESTIONS

Asked at every gate, answered with evidence or not at all

- What rung has this event earned - and who owns its file?
- Which catalogues saw this object, and how old is the newest data?
- What is the covariance calibration record for this shell?
- What are the Pc, the maximum-Pc bound and the data age, together?
- Is the counterparty alive, manoeuvrable and contactable?
- Which coordination agreement applies, and has contact been made?
- What is the latest decision time, and how was it derived?
- How many manoeuvre options exist, each priced in delta-v and fleet life?
- Has every option been secondary-screened across the full horizon?
- What cumulative risk does accepting this event spend from the shell budget?
- Does this event involve a crewed vehicle or a high-consequence object?
- Who holds decision authority at this risk level, and are they rested?
- Has any dissent been recorded in the file?
- Is any autonomous action proposed, and inside which signed envelope?
- When was the envelope last re-authorised, and on what evidence?
- What tracking has been tasked to shrink this uncertainty?
- How will the outcome be verified after TCA or after the burn?
- What does our own derelict register say this quarter?
- What is the disposal reliability of each spacecraft nearing its window - individually?
- Which passivation evidence is on file, and which is missing?
- What did the last breakup drill get wrong?
- What is the casualty expectation for the next planned re-entry?
- Which licence conditions fall due this year, per spacecraft?
- Are we sharing ephemerides at the fidelity we demand from others?
- When did a vendor safety model last change, and what validation came with it?
- What is our audited screening recall, per shell, this quarter?
- What is the cost per verified resolved conjunction, with recall printed beside it?
- How much avoidance delta-v did the fleet spend this year, and on events later judged necessary?
- Who halted an automation this year - and if nobody, is halt authority real?
- If a collision happened tonight, what would our event files prove by morning?

A question answered with a dashboard is not answered; with the event files and the calibration record, it is.

EXECUTIVE LENS

Question 30 is the one to ask in the room, out loud.

MISSION LENS

Unanswered questions are findings, not homework.

OPERATIONS LENS

Evidence or not at all - the gate rule for every volume in this series.

Six gates, each with the evidence that opens it

Gate	Decision	Evidence required
CD0 - Scope	Define shells, event classes and refusals	Shell budgets; standing refusals and decision-authority table signed
CD1 - Prove	Catalogue and covariance stand up	Calibration report per shell; audited screening recall against reference
CD2 - Pilot	Shadow assessment on live events	Shadow-mode comparison vs human desk; miss and dissent logs complete
CD3 - Integrate	Chain drives the safety desk	Event files, escalation clocks, coordination register live; secondary screening mandatory in tooling
CD4 - License	Operational reliance at earned rungs	Per-shell licences; disposal watch per spacecraft; halt and restart drilled
CD5 - Scale	Expand shells, fleet and services	Quarterly recall and calibration audits funded; unit cost verified with recall beside it

First 90 days

- **Days 1-30.** Build the event-file and derelict registers; publish the decision-authority table; freeze threshold changes outside the safety case.
- **Days 31-60.** Run covariance calibration and reference screening audits per shell; make secondary screening mandatory in the planning tool.
- **Days 61-90.** Drill a breakup surge and a halt-and-restart; hold the first CD4 gate on evidence.

EXECUTIVE LENS

Gates open on evidence, not on enthusiasm.

MISSION LENS

The 90 days end with a shorter claim and a longer memory.

OPERATIONS LENS

CD5 expands shells; it never widens the autonomy fence by itself.

What an auditor, a regulator or an inquiry will eventually ask for

The evidence pack, per operator

- Shell budgets and cumulative accepted-risk accounts, per year.
- Catalogue provenance, coverage and staleness reporting per shell.
- Covariance calibration curves and re-calibration history.
- Screening recall audits against reference screening, with every miss investigated.
- Event files for all action-threshold events, including dissents and non-decisions.
- Coordination agreements register and counterparty reliability record.
- Manoeuvre plans with secondary screening, as-flown grading and delta-v accounts.
- Disposal and passivation evidence per spacecraft; the derelict register.
- Breakup drill records and real-event after-action reviews.

Sector annex - where the emphasis moves

Sector	Emphasis
Mega-constellations	Screening scale, coordination automation under human commitment, shell budgets
Crewed operations	Station protocols override all economies; earliest moves; zero autonomous exceptions
Geostationary operators	Graveyard verification, long-horizon drift management, neighbour coordination
Government and defence	Protected sharing channels, sovereign catalogue baseline, attribution discipline
Launch and re-entry	Casualty expectation, demise verification, window communication honesty
Removal and servicing ventures	Consent and title, characterisation evidence, abort design at every range

EXECUTIVE LENS

Assemble the pack before anyone asks; afterwards is too late and it shows.

MISSION LENS

The annex moves emphasis, never the standard.

OPERATIONS LENS

The derelict register is the sector annex nobody wants and everybody needs.

EIGHT FILL-IN TEMPLATES

The licence, made paperwork

Template 1 - Event Licence Card

Event / objects: ___ | Shell: ___ | Rung: ___ | Pc / max Pc / data age: ___ | Decision owner: ___ | Deadline: ___

Template 2 - Covariance Calibration Log

Shell: ___ | Period: ___ | Objects scored: ___ | Containment vs stated sigma: ___ | Recalibration applied: ___ | Signed: ___

Template 3 - Coordination Record

Counterparty: ___ | Agreement applied: ___ | Contact made / time: ___ | Who moves: ___ | Commitment locked: ___ | File link: ___

Template 4 - Manoeuvre Plan and Secondary Check

Options and delta-v prices: ___ | Secondary screening result: ___ | Return plan: ___ | Approved by / time: ___ | As-flown grade: ___

Template 5 - Autonomous Envelope Authorisation

Shell / object classes: ___ | Thresholds: ___ | Delta-v caps: ___ | Exclusions: ___ | Expiry: ___ | Two signatures: ___

Template 6 - Disposal and Passivation Record

Spacecraft: ___ | Method / window: ___ | Reliability estimate: ___ | Passivation evidence: ___ | Verified by tracking: ___

Template 7 - Breakup Response Log

Declared / by: ___ | Cloud model version: ___ | Re-screening priority: ___ | Notifications: ___ | After-action due: ___

Template 8 - Verified Safety Workbook

Period: ___ | Fully loaded chain cost: ___ | Verified resolved conjunctions: ___ | Recall: ___ | Cost per verified resolved conjunction: ___ | Claimed / released / banked: ___

Events, calibration, coordination, manoeuvres, envelopes, disposal, breakups and cost - each short enough to complete at 3 a.m., which is when several of them will be used. Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per object, per shell, per operator and per manoeuvre window.
- A predicted conjunction is not a collision avoided.
- No Pc without covariance quality, data age and the maximum-Pc bound.
- Thresholds are signed risk appetite, never queue management.
- Screening is judged on audited recall, not alerts suppressed.
- Every manoeuvre is secondary-screened and graded as flown.
- Coordination commitments lock; both-move is designed out.
- A human decides every manoeuvre; autonomy executes prior decisions only, inside its fence.
- Events close on verification, not on silence.
- Cumulative risk is budgeted per shell, per year.
- Disposal is verified per spacecraft; fleet averages are refused.
- Passivation is evidenced, never promised.
- Your derelicts count against you until they decay or are removed.
- Share your best ephemerides at the fidelity you demand.
- Vendor safety models change only under change control.
- Casualty expectation is owed to people who never signed anything.
- Removal follows consent and title, then characterisation, then rendezvous.
- Cost per verified resolved conjunction - with recall printed beside it.
- Anyone can halt the automation; restart takes evidence and two signatures.
- A smaller true licence beats a grand claim that folds.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built
4	Earth Observation and Data	EO0-EO5	Built
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Built - this volume
6	Ground Systems and Mission Software	GS0-GS5	Next
7	Lunar and Deep-Space Operations	LD0-LD5	Planned
8	Space Security and Resilience	SR0-SR5	Planned
9	Space Economics and Fleet Finance	EF0-EF5	Planned
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 6, Ground Systems and Mission Software, is next. The series closes at ten with the Space Capstone and Assurance.